SFRI CLSA Case Analysis

Course: Sustainable Finance and Responsible Investment

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1. Introduction & Case Content

The integration of Environmental, Social, and Governance (ESG) considerations into investment decision-making has become increasingly important in contemporary financial analysis, particularly in sectors with significant environmental and regulatory exposure. Traditional valuation frameworks, which focus primarily on financial performance and capital structure, are now being supplemented with sustainability considerations to better capture long-term risks and value drivers. This shift reflects the growing recognition that ESG-related factors can materially influence cash flows, cost structures, regulatory compliance, and ultimately, firm valuation.

The cement industry represents one of the most carbon-intensive segments of the manufacturing sector and is therefore at the centre of global sustainability and decarbonisation efforts. Cement producers face increasing pressure from regulators, investors, and other stakeholders to reduce carbon emissions, improve energy efficiency, and adopt more sustainable operating practices. These pressures often require substantial upfront investments in cleaner technologies, alternative fuels, and compliance mechanisms, which may adversely impact short-term financial performance but are expected to enhance long-term business resilience and regulatory alignment.

Dalmia Bharat Limited operates within this challenging environment and has been identified as a firm that is actively engaging with sustainability initiatives while continuing to pursue growth and operational expansion. The CLSA case provides detailed financial information on the company and offers an appropriate context to examine how ESG-related investments and compliance costs interact with traditional valuation outcomes. Given the capital-intensive nature of the cement industry and the long gestation period associated with capacity expansion and technological upgrades, the firm presents a suitable case for analysing the trade-off between short-term financial impact and long-term value creation.

The objective of this study is to estimate the intrinsic equity value of Dalmia Bharat Limited by applying a Discounted Cash Flow framework that explicitly incorporates ESG considerations. The analysis adopts a Free Cash Flow to the Firm based approach, allowing for a comprehensive assessment of operating performance independent of financing decisions. ESG integration is achieved by modelling the impact of sustainability-related investments through adjustments to projected cash flows, rather than altering discount rates or growth

assumptions. This approach ensures consistency with valuation theory while enabling a structured evaluation of ESG-related financial implications.

By examining multiple ESG impact scenarios and conducting sensitivity analysis on key valuation parameters, the study aims to highlight how sustainability considerations influence firm value under varying assumptions. The findings of this analysis are intended to demonstrate that while ESG investments may impose short-term financial costs, particularly in a carbon-intensive sector, they play a critical role in long-term value preservation, regulatory compliance, and responsible capital allocation. As such, the case provides a practical illustration of how sustainable finance principles can be meaningfully integrated into equity valuation.

2. Methodology & Valuation Framework

The valuation of Dalmia Bharat Limited in this study is conducted using a Discounted Cash Flow framework based on Free Cash Flow to the Firm (FCFF). The FCFF approach is considered appropriate given the capital-intensive nature of the cement industry and the firm's evolving capital structure over time. By focusing on cash flows available to all providers of capital, the methodology allows for an assessment of the company's operating performance independent of financing decisions, thereby providing a cleaner and more comprehensive measure of intrinsic firm value.

The FCFF-based valuation framework involves projecting operating cash flows over a defined forecast horizon and discounting these cash flows at the firm's Weighted Average Cost of Capital. The forecast period is designed to capture the expected operational trajectory of the firm, including the effects of capacity expansion, cost structures, and sustainability-related investments. While operating projections are extended beyond the initial years to reflect the long-term nature of the business, the valuation exercise itself is restricted to a ten-year horizon, consistent with the assignment guidelines. Cash flows beyond this period are captured through a terminal value, which represents the continuing value of the firm under stable growth assumptions.

Revenue projections are anchored to a macroeconomic growth proxy, reflecting the linkage between cement demand, infrastructure development, and overall economic activity. Operating profitability is estimated using a stable EBIT margin derived from historical performance, ensuring that projections remain grounded in observed operating efficiency rather than optimistic expansion assumptions. This approach avoids excessive sensitivity to short-term fluctuations while maintaining consistency with the firm's demonstrated operating profile.

Capital expenditure, depreciation, and working capital assumptions are incorporated in line with the modelling constraints specified in the assignment. Capital expenditure is assumed to grow at a fixed rate based on prior-year levels, while depreciation is estimated using a historical average approach rather than a detailed asset roll-forward schedule. Changes in working capital are projected using a constant growth assumption applied to prior-year changes. These simplifying assumptions are employed to maintain transparency and focus on core valuation drivers, while remaining faithful to the information available in the case study.

The discount rate used in the valuation is the Weighted Average Cost of Capital, calculated using a combination of the cost of equity and after-tax cost of debt. The cost of equity is

estimated using the Capital Asset Pricing Model, incorporating a risk-free rate based on government bond yields, a market risk premium derived from long-term market returns, and a firm-specific beta. The cost of debt is derived from the company's historical financials and adjusted for taxation. Capital structure weights are based on the firm's balance sheet composition as provided in the case data. The resulting WACC is applied consistently across all valuation scenarios to ensure comparability.

Environmental, Social, and Governance considerations are integrated into the valuation framework through explicit adjustments to projected cash flows rather than through changes to discount rates or growth assumptions. This approach reflects the view that ESG-related initiatives primarily affect operating costs and investment requirements in the near to medium term, while avoiding subjective alterations to systematic risk parameters. Three ESG impact scenarios are modelled, representing varying degrees of sustainability-related investment intensity. Each scenario applies a predefined reduction to FCFF, allowing for a structured comparison of valuation outcomes under differing ESG assumptions.

In addition to the base valuation and ESG scenarios, sensitivity analysis is conducted to assess the robustness of the valuation to changes in key assumptions. Variations in the discount rate and growth rate are examined to identify the parameters to which firm value is most sensitive. This analysis provides insight into the relative importance of capital costs and growth expectations in determining long-term value and supports a more nuanced interpretation of the valuation results.

Overall, the methodology adopted in this study balances theoretical rigour with practical modelling constraints. It ensures consistency with established valuation principles while allowing for meaningful integration of ESG considerations, thereby aligning the analysis with the objectives of sustainable finance and responsible investment.

3. Data Sources & Key Assumptions

The valuation analysis is based entirely on financial and market data provided in the CLSA case materials, supplemented by publicly available macroeconomic and capital market information where explicitly required by the assignment guidelines. No external financial databases or third-party estimates beyond the specified sources have been used, ensuring consistency with the case framework and transparency in modelling.

Historical financial data for Dalmia Bharat Limited, including income statement and balance sheet information for the period 2019 to 2022, forms the foundation for the operating forecasts. These historical figures are used to derive key operating parameters such as revenue trends, operating profitability, capital expenditure patterns, depreciation levels, and working capital behaviour. The use of multiple years of historical data allows the assumptions to reflect the firm's underlying operating characteristics rather than short-term volatility.

Capital market inputs used in the valuation are derived from credible and widely accepted sources. The risk-free rate is based on the yield of Indian government securities, reflecting prevailing long-term sovereign borrowing costs. Market returns are calculated using historical NIFTY 50 index data over a ten-year period, providing a representative estimate of long-term equity market performance. The firm's equity beta is obtained from publicly available market data and reflects the systematic risk associated with Dalmia Bharat Limited relative to the broader market.

Particulars	Values
Risk-free rate	6.607%
Market return	15.05%
Beta	0.83
Cost of equity	13.62%
Cost of debt (after tax)	4.55%
Equity Weight	0.83
Debt Weight	0.17
WACC	12.07%
Terminal growth rate	6.50%

Table 1: Key Valuation Inputs

The cost of equity is estimated using the Capital Asset Pricing Model, incorporating the risk-free rate, market risk premium, and firm-specific beta. The cost of debt is derived from the company's historical interest expense and total debt as reported in the case financials and is adjusted for the applicable corporate tax rate to arrive at the after-tax cost of debt. Capital structure weights are calculated using the relative proportions of debt and equity from the balance sheet data. These inputs are combined to compute the Weighted Average Cost of Capital, which is applied consistently across all valuation scenarios.

Revenue growth assumptions are linked to India's long-term economic growth prospects, recognising the close relationship between cement demand, infrastructure development, and macroeconomic activity. A constant growth rate is applied across the forecast period to ensure stability and to avoid introducing cyclical or speculative assumptions into the model. Operating profitability is estimated using an average EBIT margin calculated from historical performance over the period 2019 to 2022. This approach assumes that the firm maintains a broadly stable operating structure over the forecast horizon.

Capital expenditure is projected using a fixed year-on-year growth rate based on the previous year's capex, in line with the modelling constraints specified in the assignment. Depreciation expense is estimated as the average of the most recent four historical years, avoiding the need for a detailed fixed asset roll-forward schedule. Changes in working capital are projected using a constant growth assumption applied to prior-year changes, reflecting gradual scaling of operations without introducing aggressive efficiency assumptions.

Assumption Category	**Parameter**	**Value Used**	**Basis / Rationale**
Revenue Growth	Annual Revenue Growth Rate	6.50%	Aligned with India's long-term GDP growth rate as reported by the World Bank; reflects macroeconomic demand for cement.
Operating Profitability	Average EBIT Margin (2019–2022)	9.46%	Calculated as the average historical EBIT margin over FY2019–FY2022 using CLSA case financials.

Capital Expenditure	Capex Growth Rate	1.00% p.a.	As per assignment guidelines; capex assumed to grow at 1% year-on-year based on previous year's capex.
Depreciation	Depreciation Method	Average of last 4 years	Depreciation estimated as the average of FY2019–FY2022 depreciation; no PPE schedule constructed as per guidelines.
Working Capital	Growth in Change in Working Capital	2.00% p.a.	Change in working capital assumed to grow at 2% annually, based on prior-year change.
Forecast Horizon	Operating Projection Period	25 years	Projections extended until FCFF turns positive to capture operational recovery.
Valuation Horizon	Explicit Valuation Period	10 years	Discounting restricted to 10 years in line with assignment requirements.
Discount Rate	WACC	12.07%	Computed using CAPM-based cost of equity and after-tax cost of debt with balance-sheet-based capital structure.
Terminal Growth	Terminal Growth Rate	6.50%	Assumed to converge to India's long-term GDP growth rate, consistent with macroeconomic conditions.
ESG Impact – Mild	FCFF Reduction	5%	Represents mild ESG-related investment and compliance impact, as specified in assignment guidelines.
ESG Impact – Moderate	FCFF Reduction	15%	Represents moderate ESG compliance and sustainability investment impact.

ESG Impact – Severe	FCFF Reduction	25%	Represents severe ESG-related cost burden and sustainability transition impact.

Table 2: Key Assumptions & Drivers

Environmental, Social, and Governance assumptions are incorporated through predefined reductions in projected Free Cash Flows to the Firm. These reductions are intended to capture the financial impact of sustainability-related investments, compliance costs, and operational adjustments required in a carbon-intensive industry. Three ESG scenarios are considered, representing mild, moderate, and severe levels of ESG impact. All other valuation parameters, including the discount rate and growth assumptions, are held constant across scenarios to ensure comparability and isolate the effect of ESG-related cash flow adjustments.

Overall, the assumptions adopted in this study are deliberately conservative, transparent, and closely aligned with historical performance and assignment guidelines. This approach ensures that the valuation outcomes are driven by clearly defined inputs and that the analysis remains robust under sensitivity testing.

4. Operating Forecast & FCFF Construction

The operating forecast for Dalmia Bharat Limited is developed by extending historical performance trends under clearly defined and conservative assumptions. The objective of the forecast is to translate the firm's expected operating performance into Free Cash Flows to the Firm, which form the basis of the valuation exercise. All projections are derived directly from the assumptions outlined in the previous section and are consistent with the modelling constraints specified in the assignment.

Revenue for the forecast period is projected using a constant growth rate aligned with India's long-term economic growth prospects. This approach reflects the close relationship between cement demand, infrastructure development, and macroeconomic activity, while avoiding the introduction of cyclical or speculative revenue patterns. The use of a stable growth assumption ensures that the forecast remains transparent and grounded in macroeconomic fundamentals rather than short-term market conditions.

Operating profitability is estimated by applying a stable EBIT margin to projected revenues. The margin is derived as the average historical EBIT margin observed over the period 2019 to 2022, based on the CLSA case financials. This assumption implies that the firm maintains a broadly consistent operating cost structure over the forecast horizon, without assuming material improvements or deterioration in efficiency. Projected EBIT therefore reflects scale-driven growth rather than margin expansion.

Capital expenditure is incorporated into the forecast using a fixed year-on-year growth rate applied to the previous year's capex. This assumption reflects the capital-intensive nature of the cement industry and captures ongoing investments required for capacity maintenance, incremental expansion, and sustainability initiatives. Depreciation expense is estimated as a constant value based on the average of the most recent four historical years, consistent with the instruction to avoid a detailed fixed asset roll-forward schedule.

Working capital requirements are modelled through projected changes in net working capital, with annual changes assumed to grow at a constant rate. This approach reflects gradual scaling of operations in line with revenue growth, without assuming aggressive improvements in working capital efficiency. The use of a simplified working capital assumption maintains consistency with the level of detail available in the case data.

Free Cash Flow to the Firm is computed by adjusting operating profits for taxation, non-cash expenses, capital expenditure, and changes in working capital. Specifically, FCFF is calculated by starting with operating profit after tax, adding back depreciation, and subtracting capital expenditure and changes in working capital. This formulation captures the cash-generating capacity of the firm's core operations before financing costs and provides a consistent basis for valuation across all scenarios.

Year	EBIT	Tax (25%)	NOPAT	Depreciation	CAPEX	Δ Working Capital	FCFF
2023	11,368.98	2,842.25	8,526.74	13,307.50	-18,786.00	-9,445.20	-6,396.96
2024	12,107.97	3,026.99	9,080.98	13,394.38	-18,973.86	-9,634.10	-6,132.61
2025	12,894.99	3,223.75	9,671.24	12,922.97	-19,163.60	-9,826.79	-6,396.18
2026	13,733.16	3,433.29	10,299.87	12,998.71	-19,355.23	-10,023.32	-6,079.98
2027	14,625.82	3,656.46	10,969.36	13,155.89	-19,548.79	-10,223.79	-5,647.32
2028	15,576.49	3,894.12	11,682.37	13,117.99	-19,744.27	-10,428.26	-5,372.18
2029	16,588.97	4,147.24	12,441.72	13,048.89	-19,941.72	-10,636.83	-5,087.93
2030	17,667.25	4,416.81	13,250.44	13,080.37	-20,141.13	-10,849.57	-4,659.90
2031	18,815.62	4,703.91	14,111.71	13,100.78	-20,342.55	-11,066.56	-4,196.61
2032	20,038.63	5,009.66	15,028.98	13,087.01	-20,545.97	-11,287.89	-3,717.88

Table 3: FCFF Build-up Base Case

Source: Excel Sheet Title – '4. FCFF BUILD-UP & SENSITIVITY'

The forecast period is extended beyond the initial years to capture the firm's transition from negative to positive free cash flows, reflecting the long-gestation nature of investments in the cement sector. However, for valuation purposes, only the first ten years of projected FCFF are discounted, in line with the assignment requirements. Cash flows beyond this period are incorporated through the terminal value, which is addressed in the subsequent section.

Overall, the operating forecast and FCFF construction translate the underlying business assumptions into a coherent cash flow profile that reflects both the operational challenges and long-term potential of the firm. This framework provides the foundation for the base case valuation and the subsequent ESG scenario analysis.

5. Base Case Valuation Results

The base case valuation of Dalmia Bharat Limited is derived by discounting the projected Free Cash Flows to the Firm over the explicit valuation horizon and adding the present value of the terminal value. The valuation reflects the firm's operating performance under the stated assumptions, without incorporating any ESG-related cash flow adjustments. This base case therefore serves as the benchmark against which the ESG scenarios are evaluated.

Valuation Component	Value
Terminal Value (₹ Lakhs)	2,42,723.63
Present Value of Terminal Value (₹ Lakhs)	14,058.82
Enterprise Value (₹ Lakhs)	10,945.10
Number of Shares Outstanding (Lakhs)	1,875.65
Intrinsic Value per Share (₹)	5.84

Table 4: Base Case Valuation Summary

Source: Excel Sheet Title – '4. FCFF BUILD-UP & SENSITIVITY'

The present value of projected FCFF over the ten-year valuation period is negative, reflecting the capital-intensive nature of the cement business and the substantial capital expenditure and working capital requirements in the initial years of the forecast. Despite positive operating profits, free cash flows remain constrained during this phase due to sustained reinvestment needs. This pattern is consistent with the long-gestation profile of large-scale manufacturing and infrastructure-linked industries.

The terminal value represents a significant proportion of the total enterprise value, highlighting the importance of long-term operating performance in the valuation of Dalmia Bharat Limited. This outcome reflects the assumption that the firm transitions to a more stable cash flow-generating phase beyond the explicit forecast period, supported by steady demand growth and a normalised operating structure. The terminal value is calculated using a constant growth approach, aligned with long-term economic growth expectations.

The enterprise value obtained from the base case valuation is converted into an equity value by adjusting for the firm's capital structure and dividing by the number of outstanding shares. The resulting intrinsic value per share represents the estimated fair value of the company's equity

under the base case assumptions. This value is not intended to be a short-term price target but rather an indication of the firm's long-term fundamental value based on projected cash flows.

Overall, the base case valuation underscores the sensitivity of firm value to long-term assumptions in capital-intensive industries. While near-term cash flows are pressured by ongoing investment requirements, the bulk of the firm's value is derived from its ability to generate sustainable cash flows over the long run. This base case provides a necessary reference point for assessing the incremental impact of ESG-related investments and compliance costs in the subsequent analysis.

6. ESG Scenario Analysis

To assess the financial implications of sustainability-related investments and compliance costs, Environmental, Social, and Governance considerations are incorporated into the valuation through scenario-based adjustments to Free Cash Flows to the Firm. This approach reflects the view that ESG initiatives primarily influence operating cash flows through increased capital expenditure, process modifications, and compliance-related costs, particularly in carbon-intensive industries such as cement manufacturing.

Scenario	Enterprise Value (₹ Lakhs)	Equity Value per Share (₹)
Base Case	10,945.10	5.84
Mild ESG Impact (5%)	10,397.85	5.54
Moderate ESG Impact (15%)	9,303.34	4.96
Severe ESG Impact (25%)	8,208.83	4.38

Table 5: ESG Scenario Valuation Comparison

Source: Excel Sheet Title – '5A. MILD ESG SCENARIO', '5B. MODERATE ESG SCENARIO', '5C. SEVERE ESG SCENARIO'

Three ESG impact scenarios are analysed to capture varying degrees of sustainability-related financial impact. These scenarios represent mild, moderate, and severe levels of ESG integration, corresponding to increasing intensity of environmental compliance and transition-related investments. In each scenario, projected FCFF is reduced by a predefined percentage, while all other valuation parameters, including the discount rate and growth assumptions, are held constant. This ensures that differences in valuation outcomes are driven solely by ESG-related cash flow impacts rather than changes in risk perception or macroeconomic assumptions.

Under the mild ESG scenario, a modest reduction in FCFF is applied to reflect limited sustainability-related investments and incremental compliance costs. The resulting valuation shows a relatively small decline in equity value compared to the base case, indicating that low-intensity ESG initiatives, while impacting near-term cash flows, do not materially impair long-term firm value. This outcome suggests that initial sustainability investments may be absorbed within the firm's existing operating and capital structure.

The moderate ESG scenario incorporates a higher reduction in FCFF, representing more substantial investments in emissions reduction, energy efficiency, and regulatory compliance. Under this scenario, the valuation impact becomes more pronounced, reflecting the cumulative effect of higher sustainability-related costs over the forecast horizon. The decline in equity value highlights the trade-off between near-term financial performance and the long-term strategic benefits of deeper ESG integration.

The severe ESG scenario represents an aggressive sustainability transition, with significant upfront and ongoing investments required to meet stringent environmental and regulatory standards. This scenario results in the largest reduction in equity value among the three cases, underscoring the financial burden associated with accelerated decarbonisation efforts. However, it is important to note that this scenario reflects a stress case and does not account for potential long-term benefits such as regulatory incentives, cost efficiencies from technological improvements, or enhanced stakeholder perception.

Across all scenarios, the analysis demonstrates that ESG integration leads to a progressive reduction in short-term valuation outcomes as the intensity of sustainability-related investments increases. At the same time, the consistent use of discount rates and growth assumptions across scenarios reinforces the interpretation that ESG initiatives primarily affect cash flow timing and magnitude rather than the firm's underlying systematic risk. These findings highlight the importance of evaluating ESG investments within a long-term strategic framework rather than through a purely short-term valuation lens.

7. Sensitivity Analysis

Sensitivity analysis is conducted to evaluate the robustness of the base case valuation to changes in key assumptions. Given the forward-looking nature of discounted cash flow valuation, the estimated equity value is inherently sensitive to variations in the discount rate and long-term growth assumptions. This analysis helps assess how changes in these critical inputs influence the intrinsic value of the firm.

			WACC		
			WACC (-0.5%)	**WACC (Base)**	**WACC (+0.5%)**
		₹ 5.84	**11.57%**	**12.07%**	**12.57%**
Growth	**Growth (-0.5%)**	**6.00%**	₹ 6.68	₹ 5.19	₹ 4.00
	Growth (Base)	**6.50%**	₹ 7.55	₹ 5.84	₹ 4.49
	Growth (+0.5%)	**7.00%**	₹ 8.60	₹ 6.61	₹ 5.08

Case Type	WACC	Growth Rate	Value per Share
Worst Case	11.07%	6.00%	₹ 4.00
Weak Case	11.07%	6.50%	₹ 4.49
Base Case	10.57%	6.50%	₹ 5.84
Strong Case	10.57%	7.00%	₹ 7.55
Best Case	10.07%	7.00%	₹ 8.60

Table 6: Sensitivity Analysis – WACC and Growth Rate

Source: Excel Sheet Title – '4. FCFF BUILD-UP & SENSITIVITY'

The analysis examines the impact of changes in the Weighted Average Cost of Capital (WACC) and the terminal growth rate, varied independently by ±0.5% around their respective base case values. These parameters are selected because they capture the combined effect of capital market conditions, firm-specific risk, and long-term economic growth expectations. All other assumptions are held constant to isolate the effect of these variables on valuation outcomes.

The results indicate that the intrinsic value of Dalmia Bharat Limited is more sensitive to changes in the discount rate than to equivalent changes in the growth rate. A reduction in WACC leads to a disproportionately higher increase in equity value per share, while an increase in WACC results in a sharp decline in valuation. This reflects the long-duration nature of the

firm's cash flows, where a significant portion of value is derived from terminal value and long-term operating performance.

Variations in the terminal growth rate also affect valuation outcomes, but the magnitude of change is comparatively moderate. This suggests that while long-term growth expectations are important, the firm's cost of capital plays a more dominant role in determining intrinsic value. The findings reinforce the importance of maintaining an efficient capital structure and managing financing costs, particularly in capital-intensive industries.

Overall, the sensitivity analysis highlights the key drivers of valuation risk and underscores the need for cautious interpretation of intrinsic value estimates. The results support the use of conservative assumptions in sustainable finance valuation exercises, especially when assessing firms operating in environmentally intensive sectors with long investment horizons.

8. Key Insights & Implications

The valuation exercise provides several important insights into the interaction between firm value, capital allocation, and sustainability considerations. By explicitly incorporating ESG-related cash flow adjustments into a Discounted Cash Flow framework, the analysis highlights how sustainability initiatives influence valuation outcomes primarily through their impact on operating cash flows rather than through changes in perceived risk or growth potential.

One key insight from the analysis is that the short-term valuation impact of ESG investments is sensitive to the intensity of sustainability-related commitments. Under mild ESG assumptions, the reduction in equity value remains relatively contained, suggesting that incremental investments in sustainability and compliance can be absorbed within the firm's existing operating structure. However, as the intensity of ESG integration increases, the valuation impact becomes more pronounced, reflecting the cumulative effect of higher compliance costs and capital requirements over the forecast horizon.

The results also underline the dominant role of the cost of capital in determining firm value. Sensitivity analysis demonstrates that relatively small changes in the discount rate led to materially larger variations in equity value compared to equivalent changes in growth assumptions. This finding is particularly relevant for capital-intensive industries such as cement, where long-dated cash flows and terminal value contribute significantly to overall valuation. From a sustainable finance perspective, this underscores the importance of maintaining financial stability and access to capital while undertaking long-term sustainability investments.

From a managerial standpoint, the analysis suggests that ESG investments should be viewed as strategic decisions rather than purely financial costs. While sustainability initiatives may exert pressure on near-term cash flows and valuation metrics, they contribute to long-term business resilience by reducing regulatory risk, enhancing environmental compliance, and strengthening stakeholder relationships. In sectors exposed to tightening environmental regulations, delayed ESG adoption may increase long-term financial risk despite short-term valuation advantages.

For investors, the findings highlight the need to adopt a long-term investment horizon when evaluating firms with significant ESG exposure. Traditional valuation models may understate the long-term benefits of sustainability initiatives if they focus narrowly on near-term cash flow

reductions. Integrating ESG considerations into valuation allows investors to better assess risk-adjusted returns and align investment decisions with responsible investment principles.

Overall, the analysis reinforces the relevance of sustainable finance frameworks in corporate valuation. By systematically incorporating ESG considerations into financial models, firms and investors can make more informed capital allocation decisions that balance financial performance with long-term sustainability objectives.

9. Conclusion

This study undertakes a valuation of Dalmia Bharat Limited using a Free Cash Flow to the Firm–based Discounted Cash Flow approach, while explicitly integrating Environmental, Social, and Governance considerations in line with the principles of sustainable finance and responsible investment. The analysis demonstrates how ESG-related factors can be incorporated into traditional valuation frameworks through adjustments to operating cash flows.

The findings indicate that ESG integration leads to a measurable impact on equity valuation in the short term, particularly under higher-intensity sustainability scenarios. However, these effects reflect the financial cost of transition and compliance rather than a deterioration in the firm's underlying economic potential. By maintaining consistent discount rates and growth assumptions across scenarios, the study isolates the cash flow implications of ESG initiatives and provides a transparent comparison of valuation outcomes.

Overall, the assignment highlights the importance of adopting a long-term perspective when evaluating firms operating in environmentally intensive industries. While ESG investments may appear value-reducing in conventional short-term valuation models, they play a critical role in ensuring regulatory alignment, operational resilience, and responsible capital allocation. The analysis reinforces the relevance of sustainable finance principles in equity valuation and supports the integration of ESG considerations into investment decision-making frameworks.